INLA, BYM and BYM2 in Pictures

Same counties, three ways to read a map

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Today

Last time we smoothed the Pennsylvania lung cancer map. Today we open the box: what INLA does, and what is inside the BYM and BYM2 models. Mostly pictures. Reading: **Moraga §5.2–5.3**.

Model	log-risk of county i	In words
Raw SIR	—	each county on its own
BYM	$\beta_0 + u_i + v_i$	overall level + a neighbours map + a noise map
BYM2	β_0+ mix of u_i and v_i	the same two maps, mixed with one knob, ϕ

```
library(sf)
library(dplyr)
library(tidyr)
library(ggplot2)
library(patchwork)
library(spdep)
library(SpatialEpi)   # pennLC
library(INLA)
```

```
data(pennLC)
d <- pennLC$data

ref <- d |>
  group_by(race, gender, age) |>
  summarise(r = sum(cases) / sum(population), .groups = "drop")   # reference rates

county <- d |>
  left_join(ref, by = c("race", "gender", "age")) |>
  group_by(county) |>
  summarise(O = sum(cases), E = sum(population * r), .groups = "drop") |>
  mutate(county = as.character(county), SIR = O / E)
```

```r
pa <- st_as_sf(pennLC$spatial.polygon) |>
  st_set_crs(4326) |>
  mutate(county = names(pennLC$spatial.polygon)) |>
  st_transform(5070) |>                                          # Albers
  left_join(county, by = "county")

risk_breaks <- c(0, 0.67, 0.83, 0.95, 1.05, 1.2, 1.5, Inf)
risk_labels <- c("< 0.67", "0.67-0.83", "0.83-0.95", "0.95-1.05",
                 "1.05-1.2", "1.2-1.5", "> 1.5")

map_risk <- function(values, title = NULL) {                     # relative risk, fixed classes
  ggplot(pa) +
    geom_sf(aes(fill = cut(values, risk_breaks, risk_labels)),
            colour = "white", linewidth = 0.15) +
    scale_fill_brewer(palette = "RdBu", direction = -1, drop = FALSE,
                      name = "Relative\nrisk") +
    labs(title = title) +
    theme_void(base_size = 11)
}

map_effect <- function(values, title = NULL, lim = max(abs(values))) {   # log scale, centred
  ggplot(pa) +
    geom_sf(aes(fill = values), colour = "white", linewidth = 0.15) +
    scale_fill_distiller(palette = "RdBu", limits = c(-lim, lim),
                         oob = scales::squish, name = "Effect\n(log scale)") +
    labs(title = title) +
    theme_void(base_size = 11)
}

tc <- function(x) tools::toTitleCase(x)                          # county names
f2 <- function(x) sprintf("%.2f", x)                             # two decimals
```

The problem in one picture

```r
p_data <- ggplot(pa) +
  geom_sf(aes(fill = E), colour = "white", linewidth = 0.15) +
  scale_fill_distiller(palette = "Greens", direction = 1, trans = "log10",
                       name = "Expected\ncases") +
  labs(title = "How much data?") +
  theme_void(base_size = 11)

map_risk(pa$SIR, "Raw SIR = O / E") + p_data
```

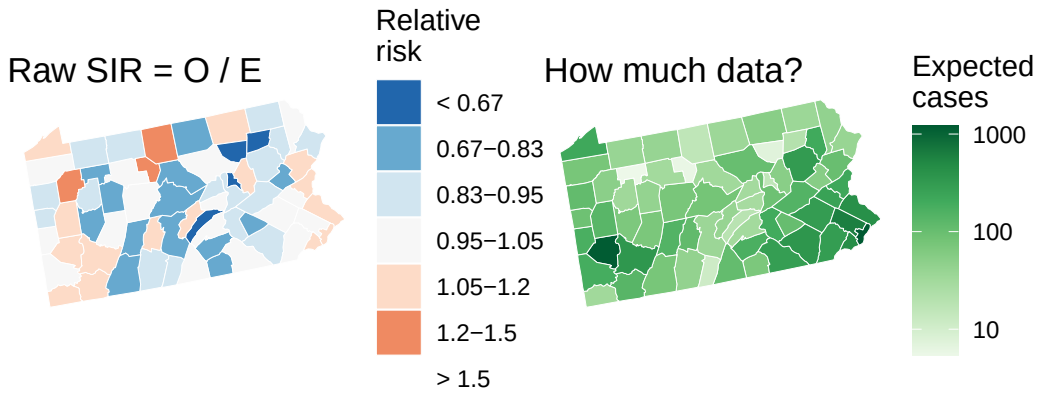


Figure 1: Left: raw SIRs. Right: how much data each county has.

The darkest reds and blues on the left tend to sit in the palest counties on the right: little data, noisy SIR. Every model today is a way to **borrow** information so these counties calm down.

INLA in two pictures

What we want: a curve, not a number

A Bayesian model returns, for each county, a whole **posterior curve**: which relative risks are plausible, and how plausible. There are two ways to get it.

```
a <- 8; b <- 7                                    # an illustrative posterior
set.seed(374)

hist_panel <- function(n) {
  ggplot(tibble(x = rgamma(n, a, b)), aes(x)) +
    geom_histogram(aes(y = after_stat(density)), binwidth = 0.1, boundary = 0,
                   fill = "grey60", colour = "white") +
    coord_cartesian(xlim = c(0, 3), ylim = c(0, 1.8)) +
    labs(title = paste0("MCMC: ", format(n, big.mark = ","), " draws"),
         x = "Relative risk", y = NULL) +
    theme_classic(base_size = 11)
}

p_inla <- ggplot(tibble(x = seq(0, 3, 0.01)), aes(x, dgamma(x, a, b))) +
```

```
  geom_area(fill = "darkgreen", alpha = 0.25) +
  geom_line(colour = "darkgreen", linewidth = 1) +
  coord_cartesian(xlim = c(0, 3), ylim = c(0, 1.8)) +
  labs(title = "INLA: the curve", x = "Relative risk", y = NULL) +
  theme_classic(base_size = 11)

hist_panel(100) + hist_panel(5000) + p_inla
```

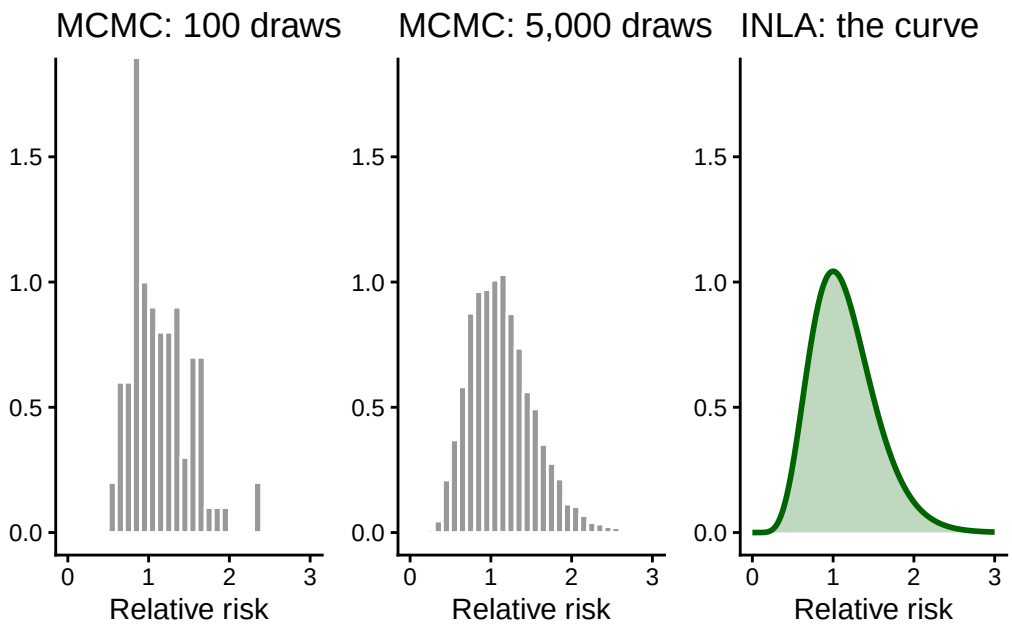


Figure 2: Same posterior, two routes. MCMC draws random values and builds a histogram; INLA computes the curve.

MCMC is like a poll: draw thousands of plausible values and count them. It works for almost any model, but it is slow, and you must check that it drew enough. **INLA** skips the drawing and computes the curve with an approximation. For disease maps it takes seconds, and its curves are usually very close to MCMC's (Rue et al. 2009).

The trick: a bell curve at the top

```
peak  <- (a - 1) / b                              # top of the curve
width <- sqrt(a - 1) / b                          # width at the top

tibble(x = seq(0, 3, 0.01),
       `True curve`     = dgamma(x, a, b),
       `Bell (Laplace)` = dnorm(x, peak, width)) |>
  pivot_longer(-x, names_to = "curve", values_to = "density") |>
```

```
ggplot(aes(x, density, colour = curve, linetype = curve)) +
geom_vline(xintercept = peak, colour = "grey60", linetype = 3) +
geom_line(linewidth = 1) +
scale_colour_manual(values = c("True curve" = "darkgreen", "Bell (Laplace)" = "darkorange")) +
scale_linetype_manual(values = c("True curve" = 1, "Bell (Laplace)" = 2)) +
labs(x = "Relative risk", y = NULL, colour = NULL, linetype = NULL) +
theme_classic(base_size = 12) +
theme(legend.position = "top")
```

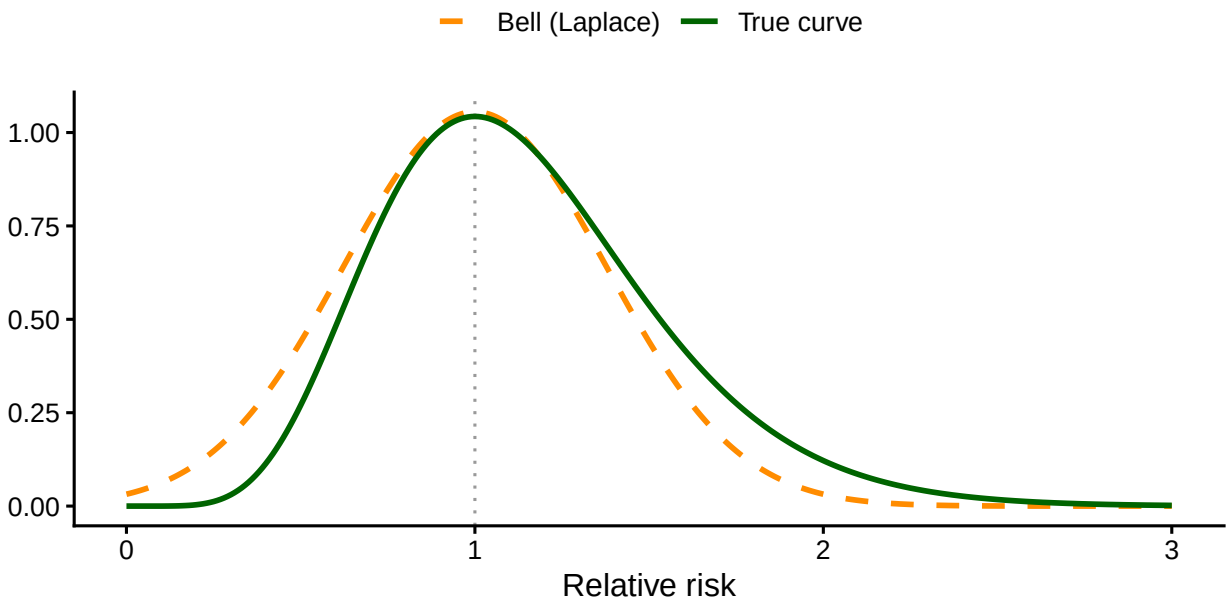


Figure 3: The Laplace approximation: a bell curve with the same peak and the same width at the top.

Close, not exact. INLA works on the log scale, where curves are closer to a bell, and then corrects the bell. The name says the rest:

Letters	Stands for	In plain words
I	Integrated	averages over the few unknown settings of the model, e.g. how smooth the map is
N	Nested	approximations inside approximations: first those settings, then each county
LA	Laplace Approximation	a bell curve fitted at the top of each curve, then refined

Key idea

You never see any of this. You write a formula; INLA returns one curve per county. Today only the `model = ...` part of the formula changes.

```
inla(O ~ 1 + f(id, model = "...", graph = g),     # overall level + county effects
     family = "poisson", E = E,                   # counts against expected cases
     data = dat)
```

Who are my neighbours?

```
nb <- poly2nb(pa, queen = TRUE)                    # queen neighbours
nb2INLA("pa.adj", nb)                              # write for INLA
g  <- inla.read.graph("pa.adj")

dat <- st_drop_geometry(pa) |>
  mutate(id = row_number())                        # one index per county
```

```
ic <- which(pa$county == "centre")                 # Centre County
xy <- unname(st_coordinates(st_point_on_surface(st_geometry(pa))))   # one point per county

links <- do.call(rbind, lapply(seq_along(nb), function(i) {
  j <- nb[[i]][nb[[i]] > i]                        # each pair once
  if (length(j) == 0) return(NULL)
  data.frame(x = xy[i, 1], y = xy[i, 2], xend = xy[j, 1], yend = xy[j, 2],
             centre = i == ic | j == ic)
}))

ggplot() +
  geom_sf(data = pa, fill = "grey95", colour = "white") +
  geom_sf(data = pa[nb[[ic]], ], fill = "#FDD49E", colour = "white") +   # neighbours
  geom_sf(data = pa[ic, ], fill = "#E6550D", colour = "white") +         # Centre
  geom_segment(data = links,
               aes(x, y, xend = xend, yend = yend, colour = centre, linewidth = centre)) +
  geom_point(data = data.frame(X = xy[, 1], Y = xy[, 2]), aes(X, Y), size = 0.8) +
  scale_colour_manual(values = c(`FALSE` = "grey55", `TRUE` = "black"), guide = "none") +
  scale_linewidth_manual(values = c(`FALSE` = 0.3, `TRUE` = 0.9), guide = "none") +
  theme_void()
```

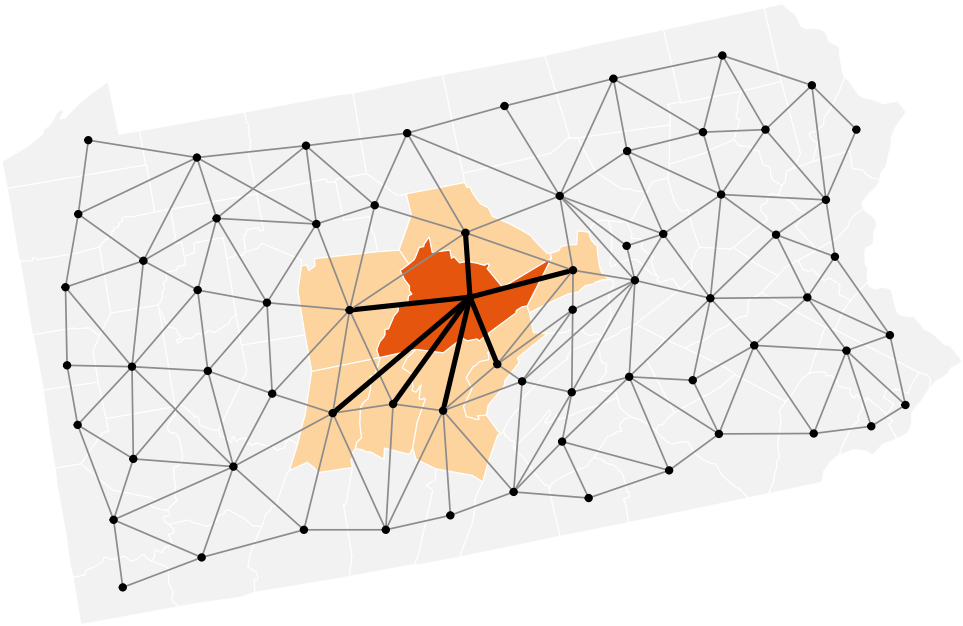


Figure 4: Each line joins two neighbours. Orange: Centre County. Light orange: its neighbours.

Centre County has 7 neighbours. In BYM and BYM2, a county borrows from the counties it is linked to, not from the whole state.

BYM: two hidden maps

Neighbours map and noise map

BYM (Besag, York and Mollié 1991) says the log-risk of county i is an overall level plus two hidden maps:

$$\log \theta_i \;=\; \beta_0 \;+\; \underbrace{u_i}_{\text{neighbours}} \;+\; \underbrace{v_i}_{\text{noise}}$$

What do these two maps look like before we see any data? Here is one random draw of each.

```
W <- nb2mat(nb, style = "B")                          # 1 = neighbours
Q <- diag(rowSums(W)) - 0.99 * W                      # nearly the ICAR (Besag) prior
set.seed(374)
u_sim <- as.vector(scale(backsolve(chol(Q), rnorm(nrow(pa)))))   # smooth map
v_sim <- as.vector(scale(rnorm(nrow(pa))))                       # pure noise
```

```
map_effect(u_sim, "u: neighbours look alike", lim = 2.5) +
  map_effect(v_sim, "v: every county for itself", lim = 2.5) +
  plot_layout(guides = "collect")
```

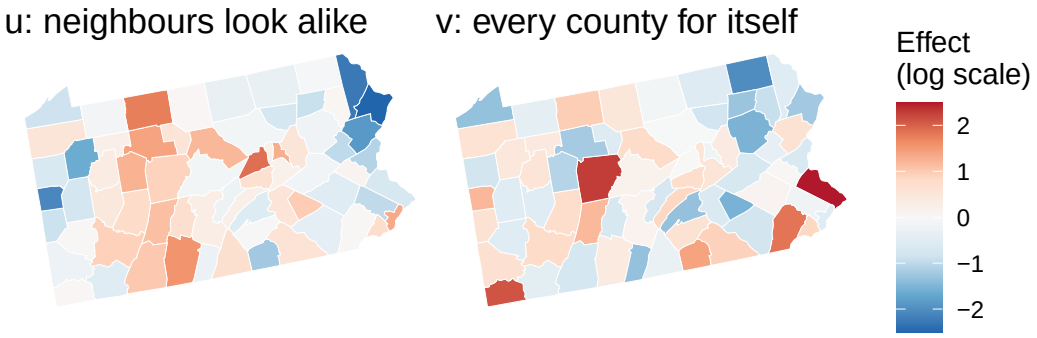


Figure 5: Left: u, a smooth map where neighbours look alike (structured). Right: v, pure county-to-county noise (unstructured). Simulated.

A real map is a mix of both. BYM lets the data decide how much of each.

Fitting BYM

```
dat$idu <- dat$id                                    # one index per f() term
dat$idv <- dat$id

m_bym <- inla(O ~ 1 + f(idu, model = "besag", graph = g) +   # u: neighbours
                      f(idv, model = "iid"),                 # v: noise
              family = "poisson", E = E, data = dat,
              control.predictor = list(compute = TRUE),
              control.compute = list(dic = TRUE, waic = TRUE))

b0          <- m_bym$summary.fixed["(Intercept)", "mean"]   # overall level
dat$u       <- m_bym$summary.random$idu$mean                # neighbours map
dat$v       <- m_bym$summary.random$idv$mean                # noise map
dat$RR_bym <- m_bym$summary.fitted.values$mean              # relative risk
```

Writing BYM as two `f()` terms shows its two pieces. INLA also offers it in one piece, `model = "bym"`.

The BYM map as an addition

```
lim <- max(abs(c(b0, dat$u, dat$v)))              # one scale for the pieces

map_effect(rep(b0, nrow(pa)), "Overall level", lim) +
  map_effect(dat$u, "+ u (neighbours)", lim) +
  map_effect(dat$v, "+ v (noise)", lim) +
  map_risk(exp(b0 + dat$u + dat$v), "= relative risk") +
  plot_layout(nrow = 1, guides = "collect") &
  theme(legend.position = "bottom")
```

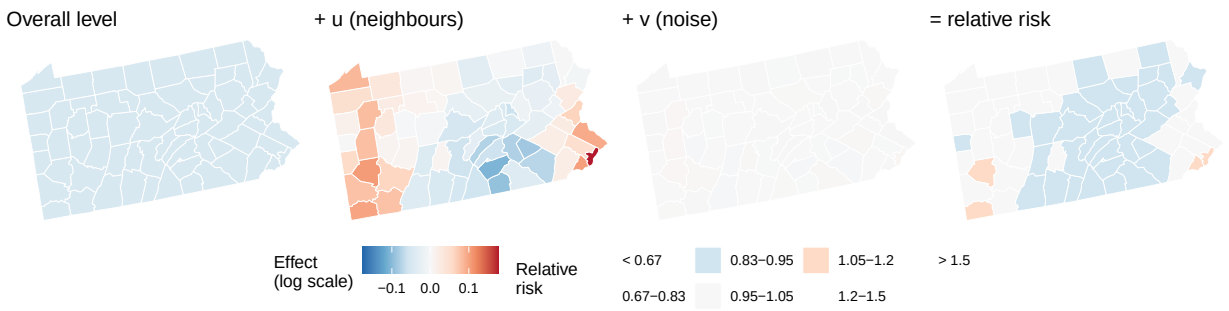


Figure 6: Pennsylvania's BYM map, piece by piece. The first three maps share one colour scale.

Read it left to right. The overall level is the same everywhere. Then each county gets a bump from its neighbourhood (u) and a bump of its own (v). Because the three pieces share one scale, the stronger colours show which hidden map carries more of Pennsylvania's pattern.

> ⚠ The catch with BYM
>
> BYM has two knobs: how much u varies and how much v varies. They are hard to read, and they compete: the data cannot easily tell which map should carry a pattern, so the two knobs are hard to estimate and to set priors for (Riebler et al. 2016). BYM2 replaces them with two knobs that mean something.

BYM2: one knob,

Turning the knob

BYM2 keeps the same two hidden maps and mixes them:

$$\log \theta_i \;=\; \beta_0 \;+\; \frac{1}{\sqrt{\tau}}\Big(\sqrt{\phi}\, u_i^{*} \;+\; \sqrt{1-\phi}\, v_i\Big)$$

u^* is u rescaled to the same typical size as v. Then τ says **how much** counties differ, and ϕ, between 0 and 1, says **how spatial** the differences are. Here is the knob, on the two simulated maps from above:

```
phis <- c(0, 0.25, 0.5, 0.75, 1)

knob <- lapply(phis, function(p) {
  map_effect(sqrt(p) * u_sim + sqrt(1 - p) * v_sim,   # BYM2 mix
             bquote(phi == .(p)), lim = 2.5) +
    theme(legend.position = "none")
})

wrap_plots(knob, nrow = 1)
```

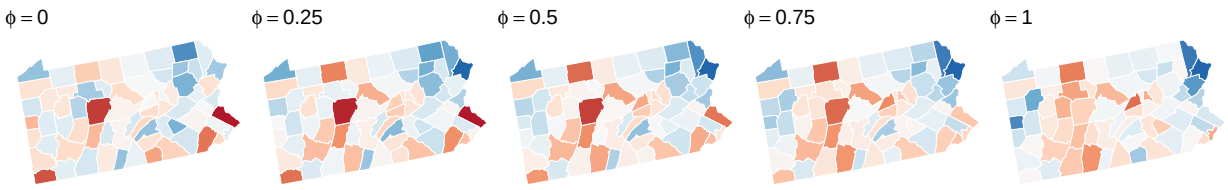


Figure 7: Mixing the same two maps with . Left: all noise. Right: all neighbours. Every panel has the same total variation.

Only the **shape** of the map changes with ϕ, not how much it varies. That is why ϕ is easy to read: the share of the variation that is spatial.

Fitting BYM2

```
m_bym2 <- inla(O ~ 1 + f(id, model = "bym2", graph = g),    # both maps in one term
               family = "poisson", E = E, data = dat,
               control.predictor = list(compute = TRUE),
               control.compute = list(dic = TRUE, waic = TRUE,
                                      return.marginals.predictor = TRUE))

dat$RR_bym2 <- m_bym2$summary.fitted.values$mean            # relative risk
round(m_bym2$summary.hyperpar[, 1:5], 2)                    #   and
```

```
                    mean    sd 0.025quant 0.5quant 0.975quant
Precision for id 132.25 55.39      56.06   121.76     269.95
Phi for id         0.53  0.23       0.10     0.54       0.93
```

We use INLA's default priors; Moraga §5.3 shows how to set your own.

Where does Pennsylvania sit on the knob?

```
phi_curve <- inla.smarginal(m_bym2$marginals.hyperpar$`Phi for id`)   # INLA's curve
phi_mean  <- m_bym2$summary.hyperpar["Phi for id", "mean"]
phi_ci    <- unlist(m_bym2$summary.hyperpar["Phi for id", c("0.025quant", "0.975quant")])

ggplot(as.data.frame(phi_curve), aes(x, y)) +
  geom_area(fill = "purple", alpha = 0.2) +
  geom_line(colour = "purple", linewidth = 1) +
  geom_vline(xintercept = phi_mean, colour = "purple", linetype = 2) +
  annotate("label", x = c(0, 1), y = 0, label = c("all noise", "all neighbours"),
           hjust = c(0, 1), vjust = 0, size = 3.5) +
  coord_cartesian(xlim = c(0, 1)) +
  labs(x = expression(paste(phi, ", share of the variation that is spatial")), y = NULL) +
  theme_classic(base_size = 12)
```

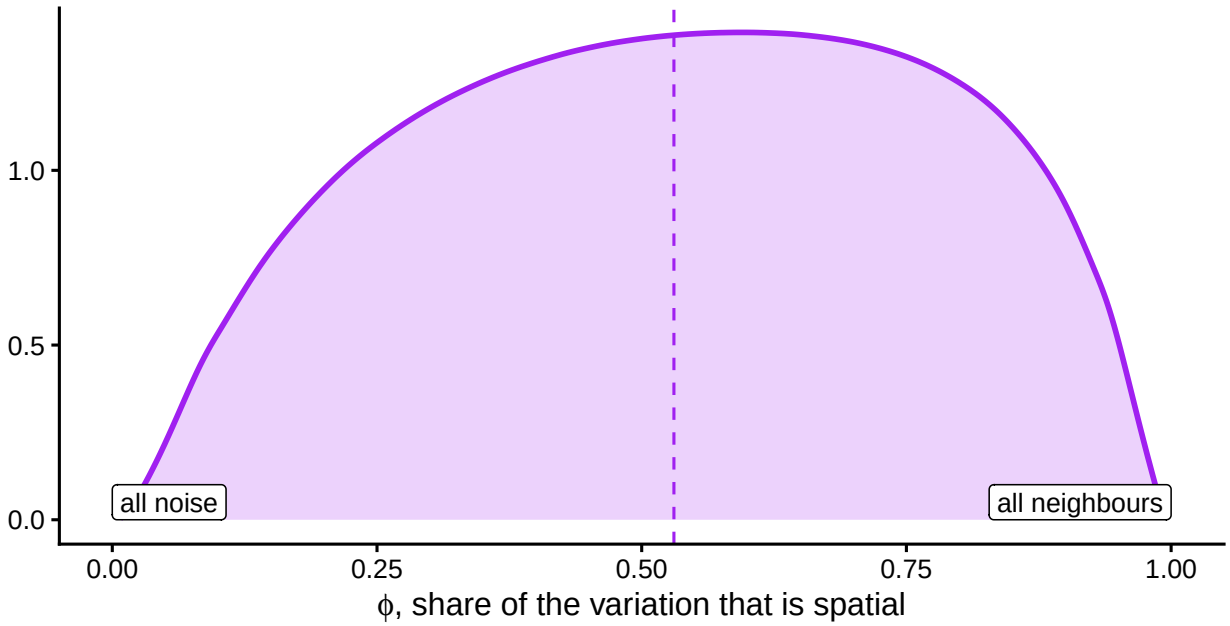


Figure 8: Posterior curve of for lung cancer in Pennsylvania. Dashed: posterior mean.

The posterior mean of ϕ is 0.53, with 95% probability between 0.10 and 0.93. In the knob figure, Pennsylvania sits closest to the panel $\phi = 0.5$. BYM could only hint at this by comparing colours; BYM2 gives it as one number, with its uncertainty.

Three maps, side by side

```
map_risk(dat$SIR, "Raw SIR") +
  map_risk(dat$RR_bym, "BYM") +
  map_risk(dat$RR_bym2, "BYM2") +
  plot_layout(nrow = 1, guides = "collect") &
  theme(legend.position = "bottom")
```

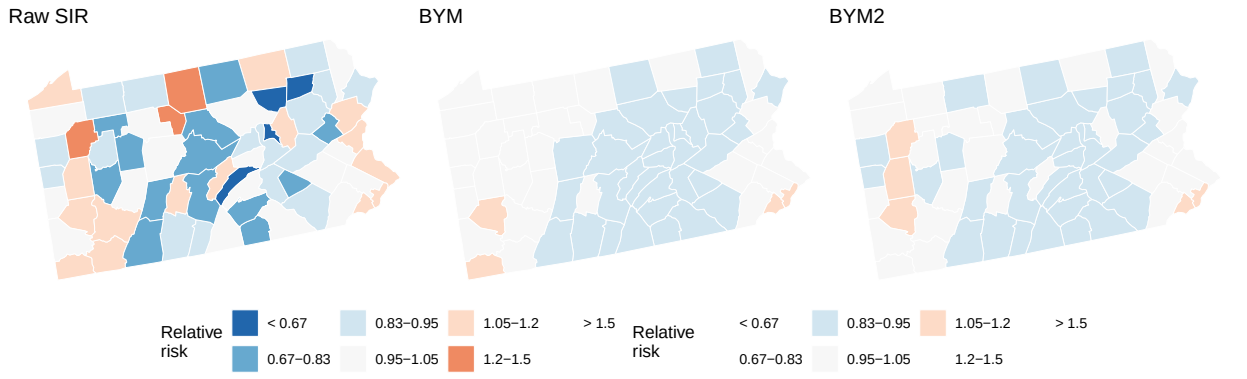


Figure 9: Raw SIR, BYM and BYM2. Same colours.

```
dat |>
  select(county, E, `Raw SIR` = SIR, BYM = RR_bym, BYM2 = RR_bym2) |>
  pivot_longer(c(`Raw SIR`, BYM, BYM2), names_to = "estimate", values_to = "risk") |>
  mutate(estimate = factor(estimate, c("Raw SIR", "BYM", "BYM2"))) |>
  ggplot(aes(estimate, risk, group = county, colour = E)) +
  geom_hline(yintercept = 1, linetype = 2) +
  geom_line(alpha = 0.7) +
  geom_point(size = 1.5) +
  scale_colour_viridis_c(option = "plasma", end = 0.9, trans = "log10",
                         name = "Expected\ncases") +
  scale_y_log10() +
  labs(x = NULL, y = "Relative risk (log scale)") +
  theme_minimal(base_size = 12)
```

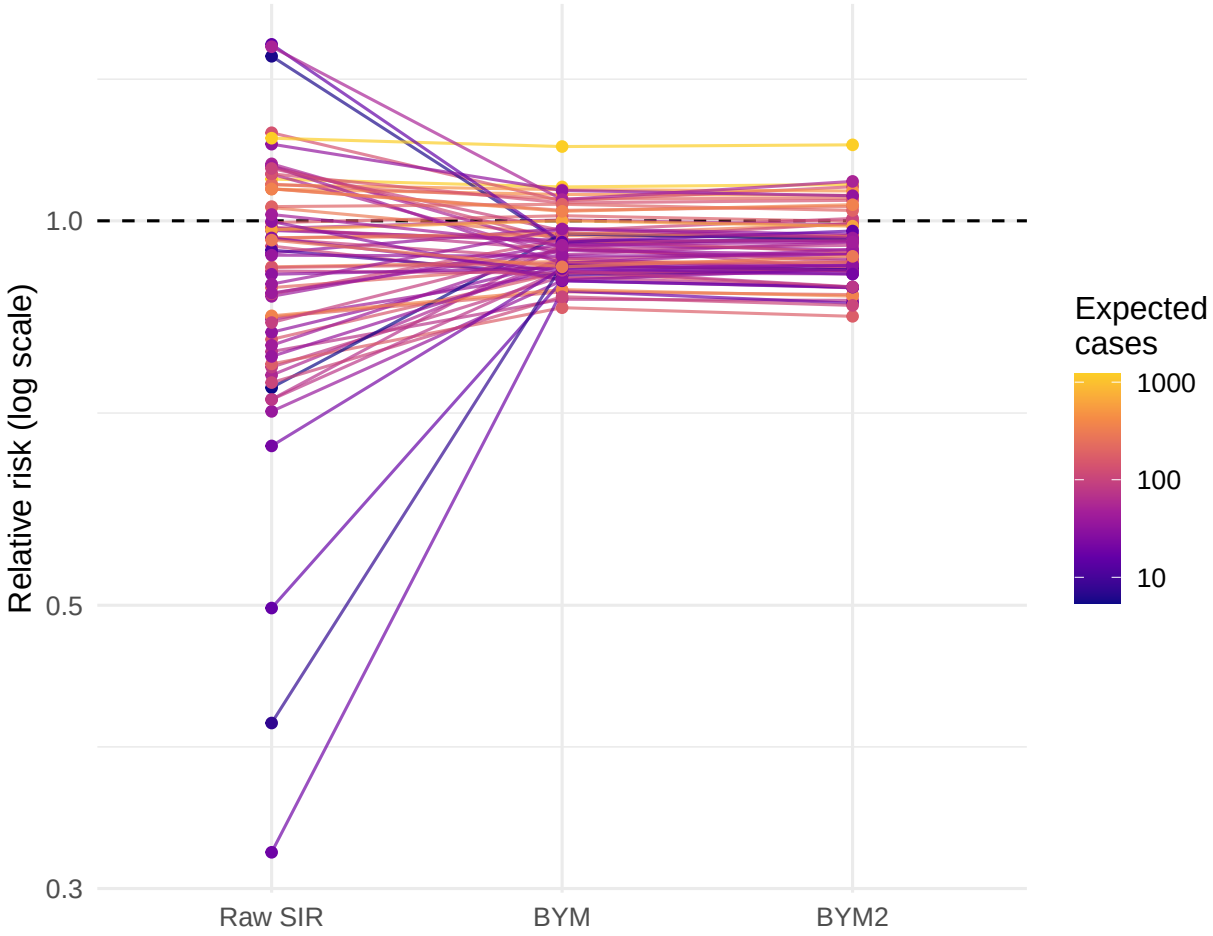


Figure 10: Where each county goes. Each line is one county; dark lines are small counties.

Small counties start far from 1 and are pulled in; big counties barely move. BYM and BYM2 nearly agree: the largest difference for any county is 0.04.

```
tibble(model = c("BYM", "BYM2"),
       DIC   = c(m_bym$dic$dic,   m_bym2$dic$dic),
       WAIC  = c(m_bym$waic$waic, m_bym2$waic$waic)) |>
  mutate(across(c(DIC, WAIC), \(x) round(x, 1)))       # lower is better
```

```
# A tibble: 2 x 3
  model   DIC  WAIC
  <chr> <dbl> <dbl>
1 BYM    515.  517.
2 BYM2   514.  513.
```

Lower is better, but a difference of a few points means little. We prefer BYM2 not for a better map but because its knobs mean something.

What we report

```
dat$p_exc <- sapply(m_bym2$marginals.fitted.values,
                    \(m) 1 - inla.pmarginal(1, m))   # P(risk > 1)
```

```
exc_cols <- c("Probably lower (< 0.2)"  = "#2c7bb6",
              "Uncertain (0.2-0.8)"     = "grey90",
              "Probably higher (> 0.8)" = "#d7191c")
exc <- cut(dat$p_exc, c(0, 0.2, 0.8, 1), labels = names(exc_cols),
           include.lowest = TRUE)

p_exc <- ggplot(pa) +
  geom_sf(aes(fill = exc), colour = "white", linewidth = 0.15) +
  scale_fill_manual(values = exc_cols, drop = FALSE, name = NULL) +
  labs(title = "How sure? P(risk > 1)") +
  theme_void(base_size = 11)

map_risk(dat$RR_bym2, "Best guess (BYM2)") + p_exc
```

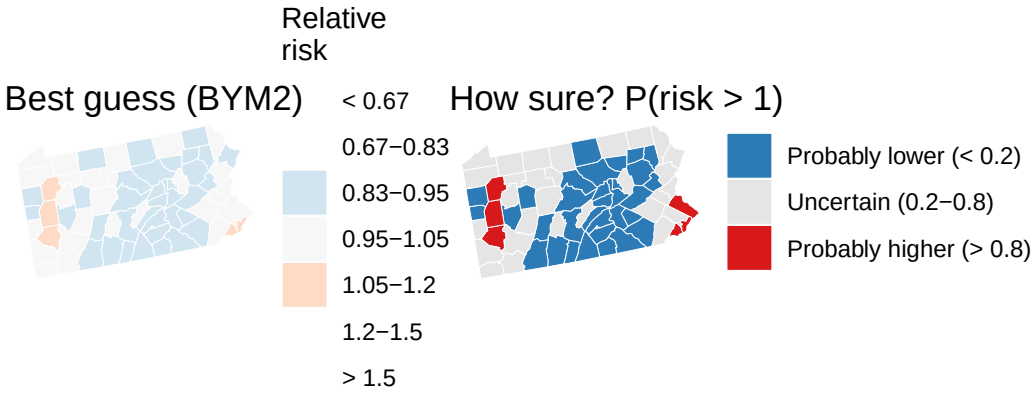


Figure 11: Left: best guess (BYM2). Right: probability that the risk is above the state average.

A good disease map comes in pairs: the best guess, and how sure we are about it.

Check yourself

1. In the BYM addition figure, which hidden map, u or v, has the stronger colours? What does that say about ϕ?
2. In the slope chart, find the line that moves the most. Is it a small or a big county? Why?
3. If ϕ were 0, what would BYM2 become? Which method from last lecture does it resemble?

Cheat sheet

```
nb <- poly2nb(pa)                                          # 1. neighbours
nb2INLA("pa.adj", nb); g <- inla.read.graph("pa.adj")
dat$id <- 1:nrow(dat)                                      # 2. one index per county

m <- inla(O ~ 1 + f(id, model = "bym2", graph = g),        # 3. model
          family = "poisson", E = E, data = dat,
          control.predictor = list(compute = TRUE))

dat$RR <- m$summary.fitted.values$mean                     # 4. relative risk
m$summary.hyperpar                                         # 5.   and
```

Words to know

Word	Plain meaning
INLA	a fast way to compute posterior curves, without drawing samples
Laplace approximation	a bell curve fitted at the top of a curve
u, structured effect	hidden map where neighbours look alike (ICAR, or Besag, prior)
v, unstructured effect	hidden map of county-to-county noise
BYM	overall level + u + v
BYM2	the same two maps, mixed by ϕ
ϕ	share of the variation between counties that is spatial
τ	precision: how little counties differ overall (large τ, small differences)

References

Besag, J., York, J., & Mollié, A. (1991). Bayesian image restoration, with two applications in spatial statistics. *Annals of the Institute of Statistical Mathematics*, 43(1), 1–20.

Moraga, P. (2019). *Geospatial health data: Modeling and visualization with R-INLA and Shiny*. Chapman & Hall/CRC, Ch. 5.

Riebler, A., Sørbye, S. H., Simpson, D., & Rue, H. (2016). An intuitive Bayesian spatial model for disease mapping that accounts for scaling. *Statistical Methods in Medical Research*, 25(4), 1145–1165.

Rue, H., Martino, S., & Chopin, N. (2009). Approximate Bayesian inference for latent Gaussian models by using integrated nested Laplace approximations. *Journal of the Royal Statistical Society: Series B*, 71(2), 319–392.